

Tinker Tailor LLM Spy: Reconstructing 'Deleted' Chats & Hijacking Sessions from Chromium LevelDB Caches

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Target Venue: Black Hat Europe Briefings

Abstract

As Large Language Model (LLM) portals become standard enterprise utilities, proprietary code and credentials are routinely processed by endpoints. When a user clicks 'Delete Chat' inside ChatGPT, Claude, or Gemini, cloud records are marked for removal, but local Chromium LevelDB storage retains write-ahead log records (.log) and uncompact Sorted String Tables (.sst/.ldb). This paper introduces **Tinker Tailor LLM Spy**, a zero-dependency forensic framework to dissect V8 serialized binary structures, parse SMI-shifted array keys, and extract deleted LLM telemetry. Official open-source repository: <https://github.com/prajwalchowdary2/tinker-tailor-llm-spy-bheu>

1. Introduction

Digital forensics investigators face a critical blind spot in client-side Generative AI telemetry. Because LevelDB is an append-only log-structured merge-tree (LSM tree), deleted records persist until compaction. Our framework dissects raw binary streams directly from disk without database lock collisions.

2. Low-Level Database & V8 Architecture

Chromium serializes IndexedDB values using V8 ValueSerializer. Lengths are encoded via Protocol Buffer varints. OneByteString (0x22) and TwoByteString (0x63 UTF-16LE) tags are decoded alongside SMI-shifted array keys (actual = encoded >> 1). Dynamic depth tracking ensures complete message reassembly.

3. Defensive Mitigations & Availability

Mitigations include GPO/MDM cache eviction policies, BitLocker encryption, and EDR path monitoring. The open-source forensic tool and whitepaper are available at: <https://github.com/prajwalchowdary2/tinker-tailor-llm-spy-bheu>